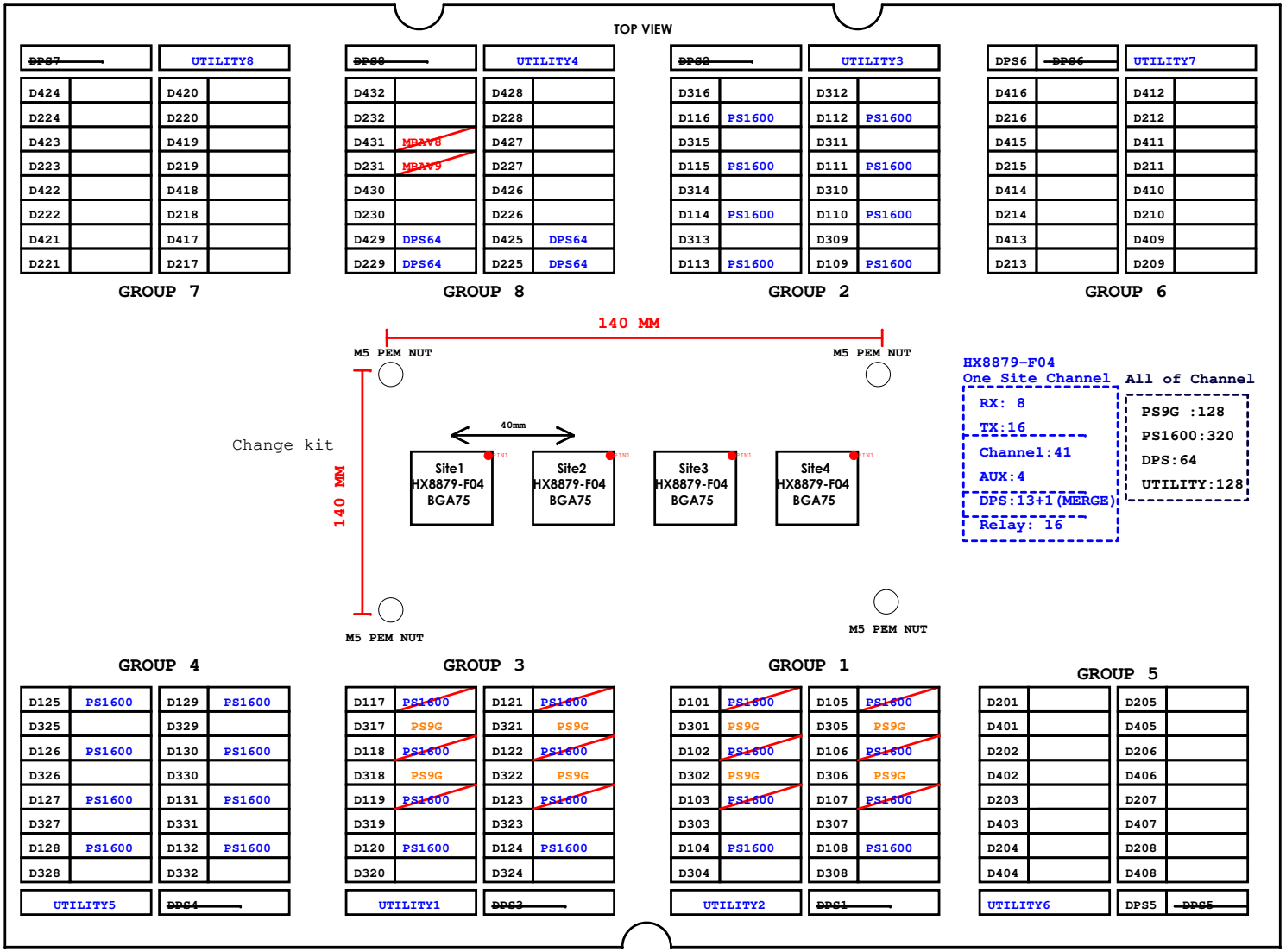
V93K FULL SIZE



Socket Drawing Follow:

Ball Out Follow:

POD Follow:

Operating temp : -40~+105

Socket M3 Nut 4*4=16

LOGO2 LOGO_HX_Himax
LOGO1 LOGO_KT_KSMT
WFBGA-10X3.5X0.636-P0.65 REF-CODE-CHIP
261103AA031 REF-CODE-SOCKET
038HX-887904K4Q-A100 REF-CODE-PCB
980HX-887904-KAG- REF-CODE-PCB
HX8879-F04 REF-CODE-CHIP

Himax

Rev.	Date	Modify History
A1	<Modify Date>	<Modify History> Initiation

KEYSTONE MICROTECH
TEL:+886-3-5509980
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E-Mail:Service@KSMT.COM.TW

Page Name	A0001-CONFIG
Product Name	980HX-887904-KAG-A10
PCB Number	038HX-887904K4Q-A100
Designer:	JACK Hsu
Ext.Num.:	6629
Approver:	<Approver>
Ext.Num.:	<Ext.>
Date:	Wednesday, November 11, 2020
Rev.	A
Size	A2
PAGE	1 of 28

History

2026 11/2 --> Modify TX to 2.7Gbps
Removed the relay from the Peripheral circuit
Keep 0402 package for NC CAP
Spare Relay Remove
J1 utilizes a DIP switch for switching

2026 11/3 --> Modify SPI VCC to EEPROM Flash VCC

2026 11/4 --> 1.Remove K12/K13/K29/K30 pull-up Relays
2.Remove TEST K49 pull-up relay
3.Remove AUX K50 Pull-down relay
4.Remove EEPROM circuit, retain K28 only.
5.Flash uses Flash Sockets
6.Power - marked current <200mA, and VDD09_CORE is < 2A
7.VDD09_CORE merge 2 CSDPS Channel
8.Modify EEPROM Flash VCC to Flasher VCC

2026 11/5 --> 1.Pin Follow EVB
2.Remove R199
3.Modify Flash_VCC to VCC_I2C_Flash_18
4.Change C14 pin name from VDDRX_AUX to VDDRX_AUX_09
5.Modify to retain AUX K50
6.REMOVE R183

PIN NUMBER	PIN NAME	PIN NUMBER	PIN NAME
A1	RESET	C3	VDD_IOB_18_POR
A10	TX3_P	C4	VDD_CORE_09
A11	TX2_P	C5	VDD_IO_18
A12	TX1_P	C6	VDD_IO_18
A13	TX0_N	C7	VDD_TXPST_09
A14	HPD	C8	VDD_CORE_09
A15	PWMI	C9	VDD_CORE_09
A2	SD_LOCK	D1	I2C_SCL
A3	XSTB	D10	RX_P0
A4	VDD_TX_09	D11	RX_P1
A5	VDD_TX_09_HDO	D12	RX_N2
A6	TX7_P	D13	RX_N3
A7	TX6_N	D14	AUX_P
A8	TX5_N	D15	GPIO10
A9	TX4_N	D2	STV2
B1	STV1	D3	GPIO7
B10	TX3_N	D4	TEST
B11	TX2_N	D5	GPIO2
B12	TX1_N	D6	GPIO3
B13	TX0_P	D7	GPIO4
B14	VDD_IOB_18_BOOST	D8	GPIO5
B15	BIST	D9	ROM_SPI_SDO
B2	CPV	E1	OE1
B3	VDD_PLL_09	E10	RX_N0
B4	VSS_CORE_09	E11	RX_N1
B5	VSS_CORE_09	E12	RX_P2
B6	TX7_N	E13	RX_P3
B7	TX6_P	E14	AUX_N
B8	TX5_P	E15	VDDRX_09
B9	TX4_P	E2	OE2
C1	I2C_SDA	E3	PWMO
C10	VDD_CORE_09	E4	GPIO8
C11	VDD_IO_18	E5	GPIO1
C12	VSS_CORE_09	E6	GPIO6
C13	VSS_CORE_09	E7	ROM_SPI_CS
C14	VDDRX_AUX_09	E8	ROM_SPI_SCK
C15	CABC_EN	E9	ROM_SPI_SDI
C2	GPIO9		

2026 11/10 --> modify Pin name
VDD_TX_09_HDO => VDD_TX_HDO_09
VDD_IOB_18_POR => VDD_IOB_POR_18
VDD_IOB_18_BOOST => VDD_IOB_BOOST_18
VSS_CORE_09 => VSS

PIN NUMBER	PIN NAME	PIN NUMBER	PIN NAME
E14	AUX_N	D2	STV2
D14	AUX_P	D4	TEST
B15	BIST	A13	TX0_N
C15	CABC_EN	B13	TX0_P
B2	CPV	B12	TX1_N
E5	GPIO1	A12	TX1_P
D15	GPIO10	B11	TX2_N
D5	GPIO2	A11	TX2_P
D6	GPIO3	B10	TX3_N
D7	GPIO4	A10	TX3_P
D8	GPIO5	A9	TX4_N
E6	GPIO6	B9	TX4_P
D3	GPIO7	A8	TX5_N
E4	GPIO8	B8	TX5_P
C2	GPIO9	A7	TX6_N
A14	HPD	B7	TX6_P
D1	I2C_SCL	B6	TX7_N
C1	I2C_SDA	A6	TX7_P
E1	OE1	C4	VDD_CORE_09[1]
E2	OE2	C8	VDD_CORE_09[2]
A15	PWMI	C9	VDD_CORE_09[3]
E3	PWMO	C10	VDD_CORE_09[4]
A1	RESET	C11	VDD_IO_18[1]
E7	ROM_SPI_CS	C5	VDD_IO_18[2]
E8	ROM_SPI_SCK	C6	VDD_IO_18[3]
E9	ROM_SPI_SDI	B14	VDD_IOB_BOOST_18
D9	ROM_SPI_SDO	C3	VDD_IOB_POR_18
E10	RX_N0	B3	VDD_PLL_09
E11	RX_N1	A4	VDD_TX_09
D12	RX_N2	A5	VDD_TX_HDO_09
D13	RX_N3	C7	VDD_TXPST_09
D10	RX_P0	E15	VDDRX_09
D11	RX_P1	C14	VDDRX_AUX_09
E12	RX_P2	B4	VSS[1]
E13	RX_P3	B5	VSS[2]
A2	SD_LOCK	C12	VSS[3]
B1	STV1	C13	VSS[4]
		A3	XSTB

Himax

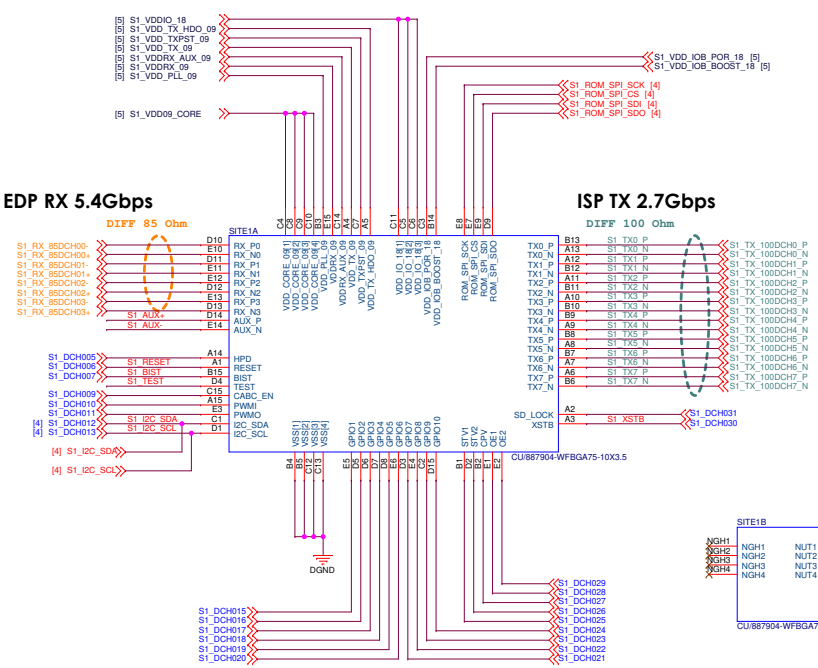
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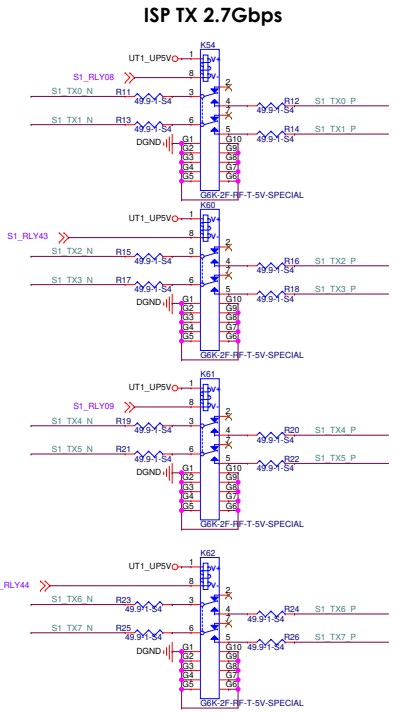
KEYSTONE MICROTECH
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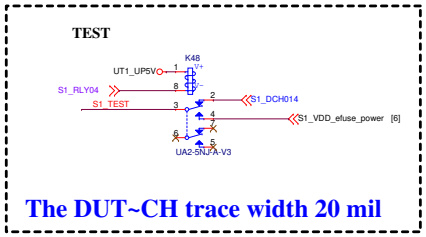
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PCB Number: 038HX-887904K4Q-A100

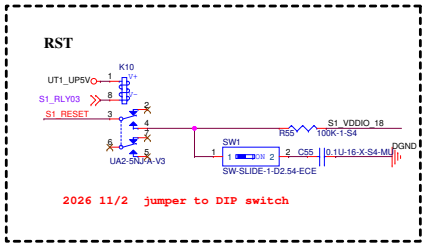
Designer: JACK Hsu
Ext.Num.: 6629
Approver: <Approver>
Ext.Num.: <Ext.>
Date: Wednesday, November 11, 2026
Rev.: A
Size: A2
PAGE: 2 of 28

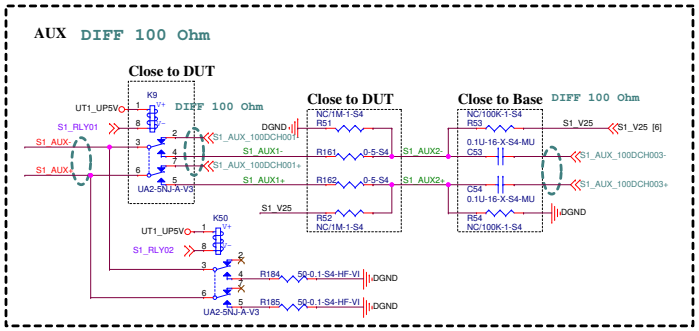
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IC PIN - RX_P0 connected to Test Channel - 85DCH-
IC PIN - RX_N0 connected toTest Channel - 85DCH+
IC PIN - RX_P1 connected to Test Channel - 85DCH-
IC PIN - RX_N1 connected to Test Channel - 85DCH+
IC PIN - RX_P2 connected to Test Channel - 85DCH-
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IC PIN - RX_P3 connected to Test Channel - 85DCH-
IC PIN - RX_N3 connected to Test Channel - 85DCH+









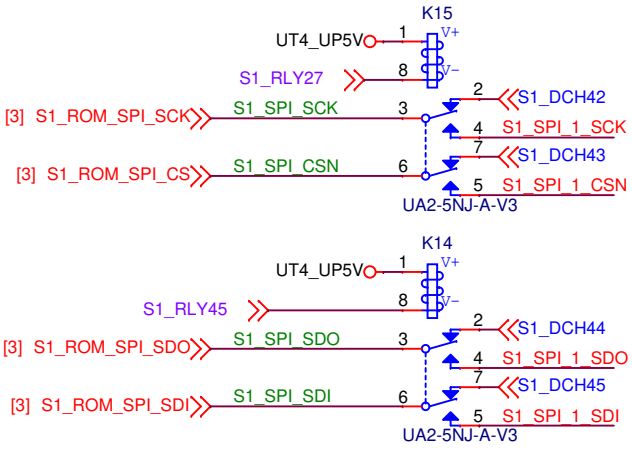


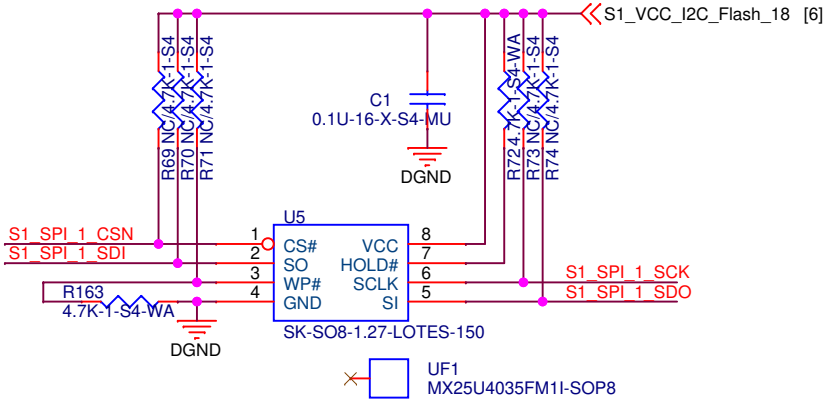
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	PCB Number	038HX-887904K4Q-A100	Date: Wednesday, November 11, 2026
	Rev. A	Size A2	PAGE 3 of 28

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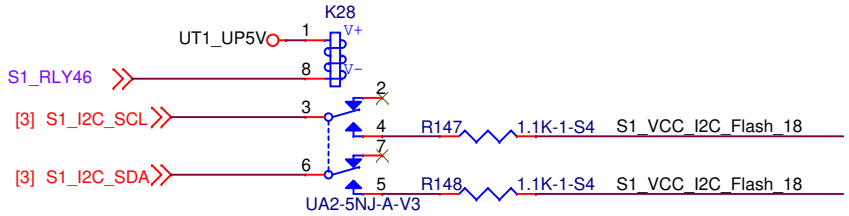
Flash uses MX25U4035FM1I-SOP8

※Please note that Flash uses the shortest path and all sites have the same length.
**Line length < 4000MIL





I2C



Rev.	Date	Modify History
A1	<Modify Date>	Initiation

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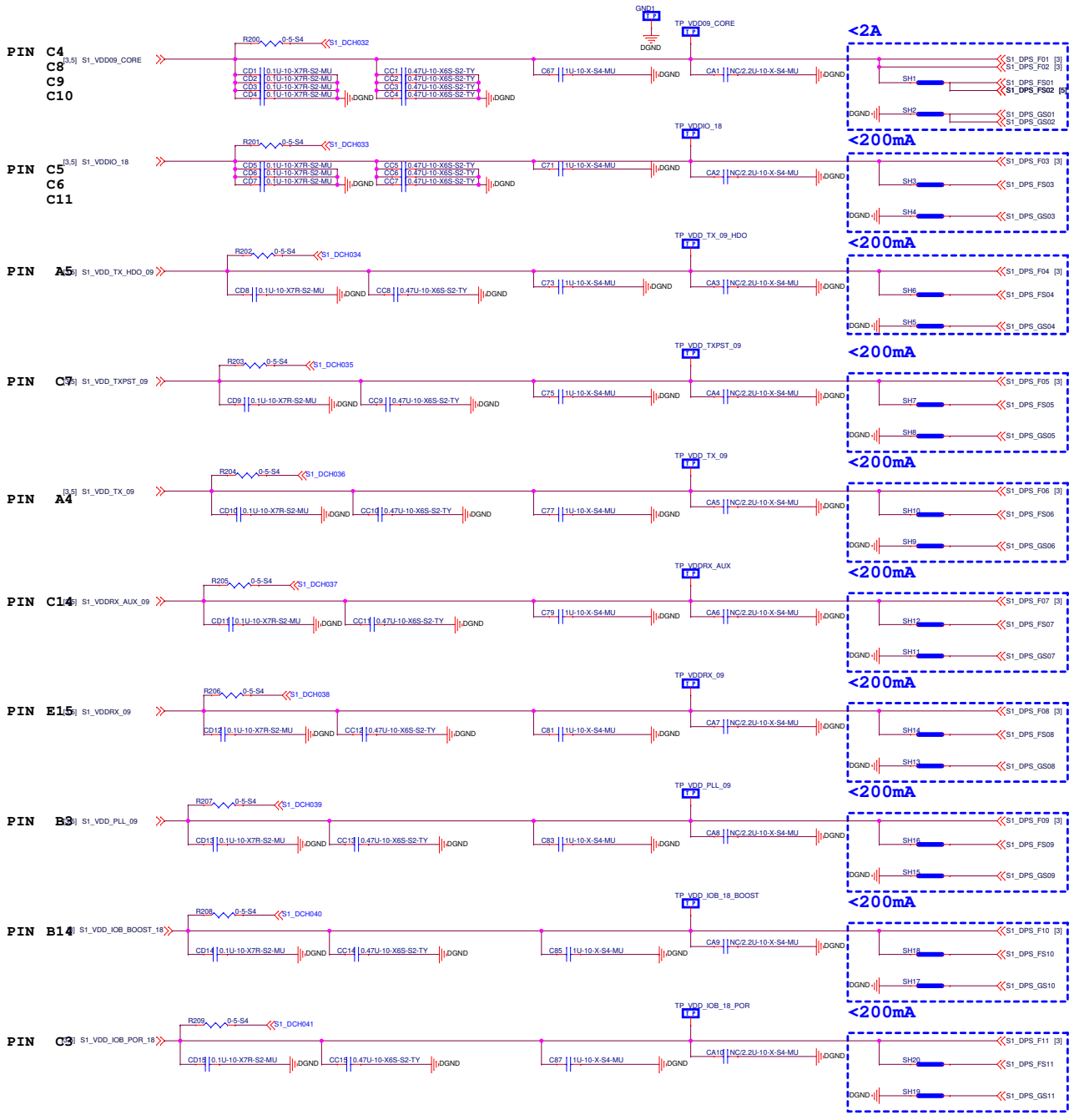
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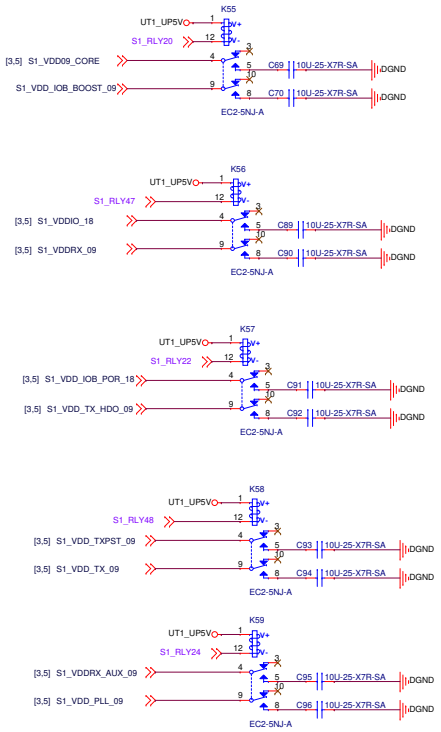
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Approver: <Approver>
Ext.Num.: <Ext.>
Date: Wednesday, November 11, 2026

Rev.	Size	PAGE
A	A4	4 of 28

CAP Reference :
CD*-->Place on TOP ; close to PIN
CC*-->Place on TOP ; close to PIN
CA*-->Place on TOP ; close to ATE

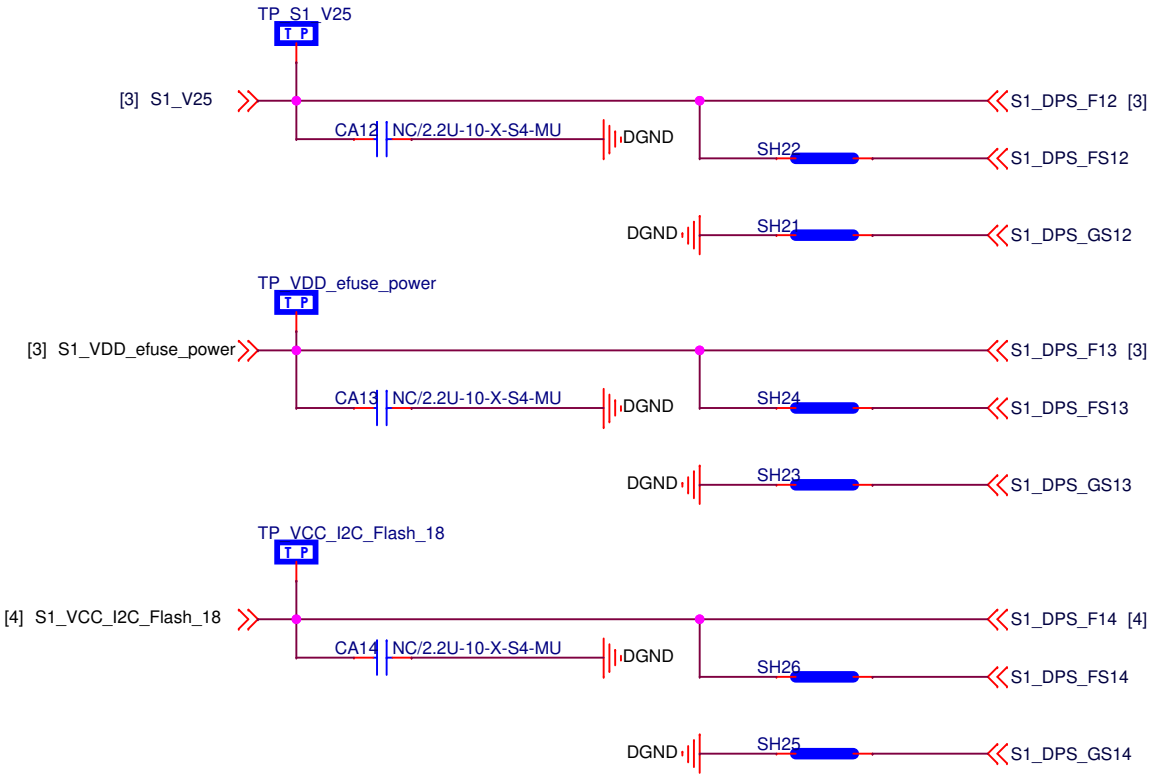
RLY close to ATE
C*--> Place on Bot ; Close RELAY





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CAP Reference :
CD*-->Place on TOP ; close to PIN
CC*-->Place on TOP ; close to PIN
CA*-->Place on TOP ; close to ATE



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Rev.	Date	Modify History

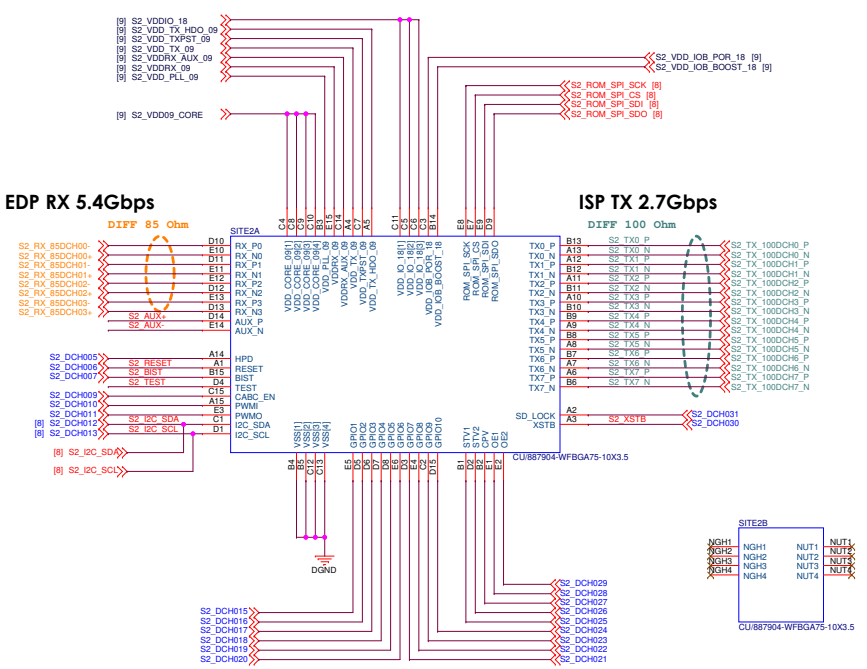
KEYSTONE MICROTECH
TEL:+886-3-5509980
FAX:+886-3-5501880
E-Mail:Service@KSMT.COM.TW

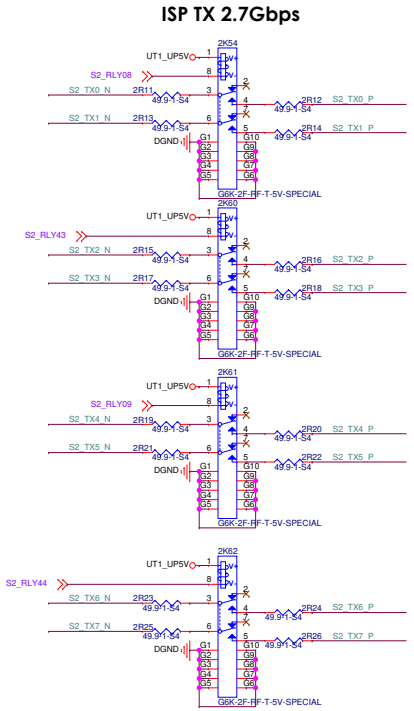
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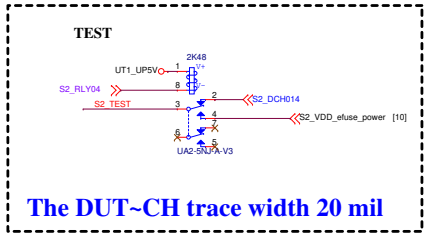
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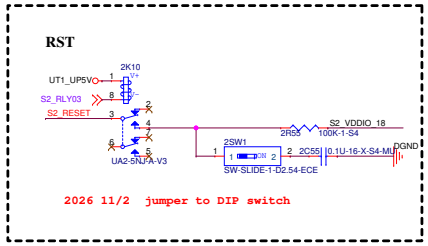
Rev.	Size	PAGE
A	A4	6 of 28

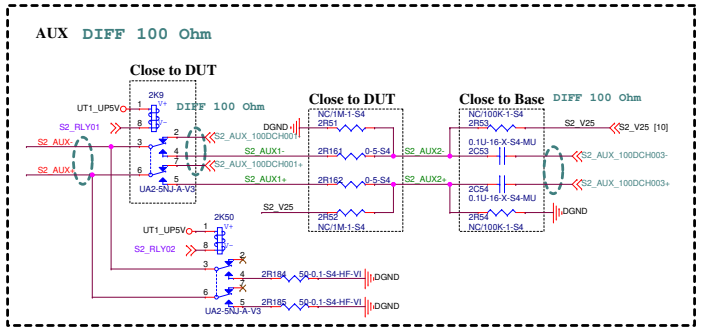
Swap P/N (P=+, N=-) to prevent RX routing from impacting trace length
IC PIN - RX_P0 connected to Test Channel - 85DCH-
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IC PIN - RX_N1 connected to Test Channel - 85DCH+
IC PIN - RX_P2 connected to Test Channel - 85DCH-
IC PIN - RX_N2 connected to Test Channel - 85DCH+
IC PIN - RX_P3 connected to Test Channel - 85DCH-
IC PIN - RX_N3 connected to Test Channel - 85DCH+











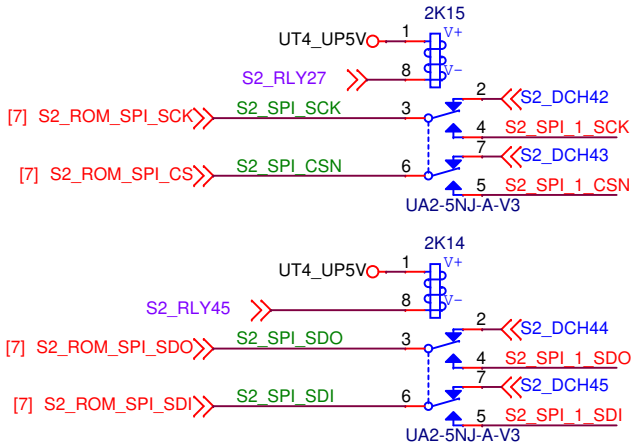
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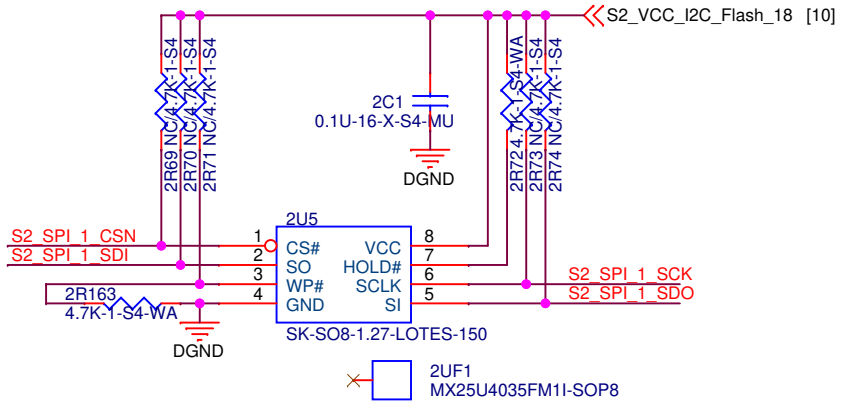
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Rev.	A
Size	A2
PAGE	7 of 28

FLASH

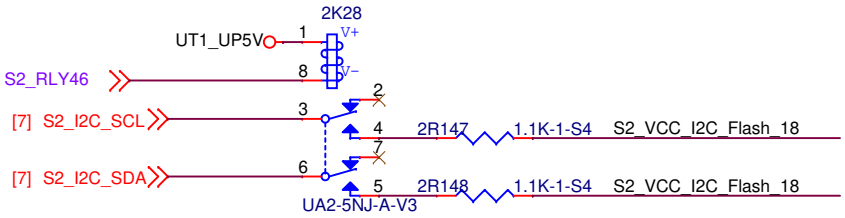


Flash uses MX25U4035FM1I-SOP8

※Please note that Flash uses the shortest path and all sites have the same length.
**Line length < 4000MIL



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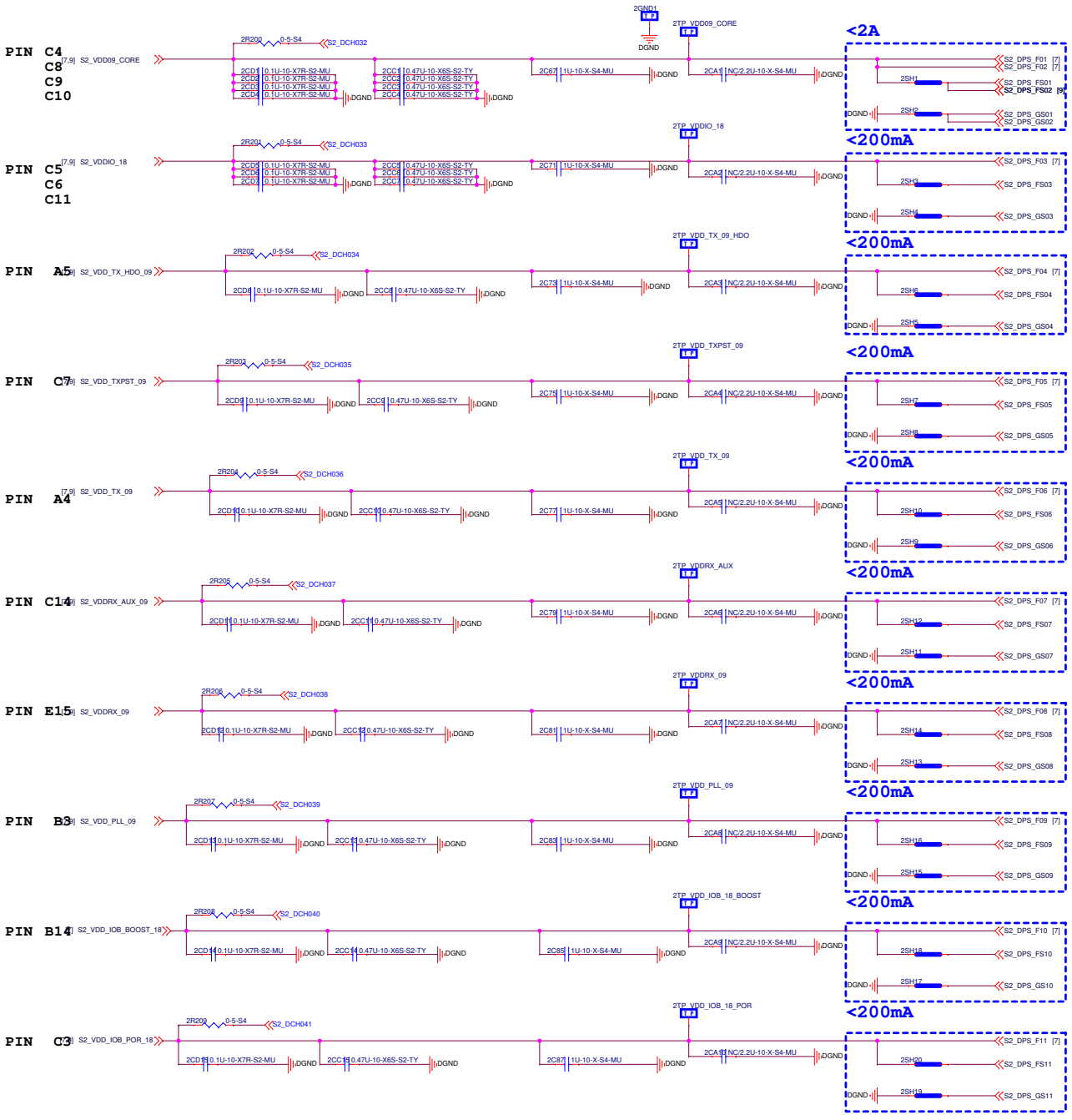
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TEL:+886-3-5509980
FAX:+886-3-5501880
E-Mail:Service@KSMT.COM.TW

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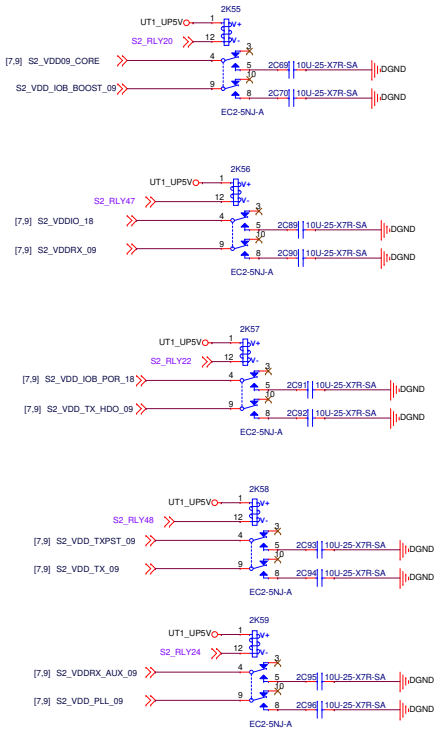
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Rev.	Size	PAGE
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CAP Reference :
CD*-->Place on TOP ; close to PIN
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CA*-->Place on TOP ; close to ATE

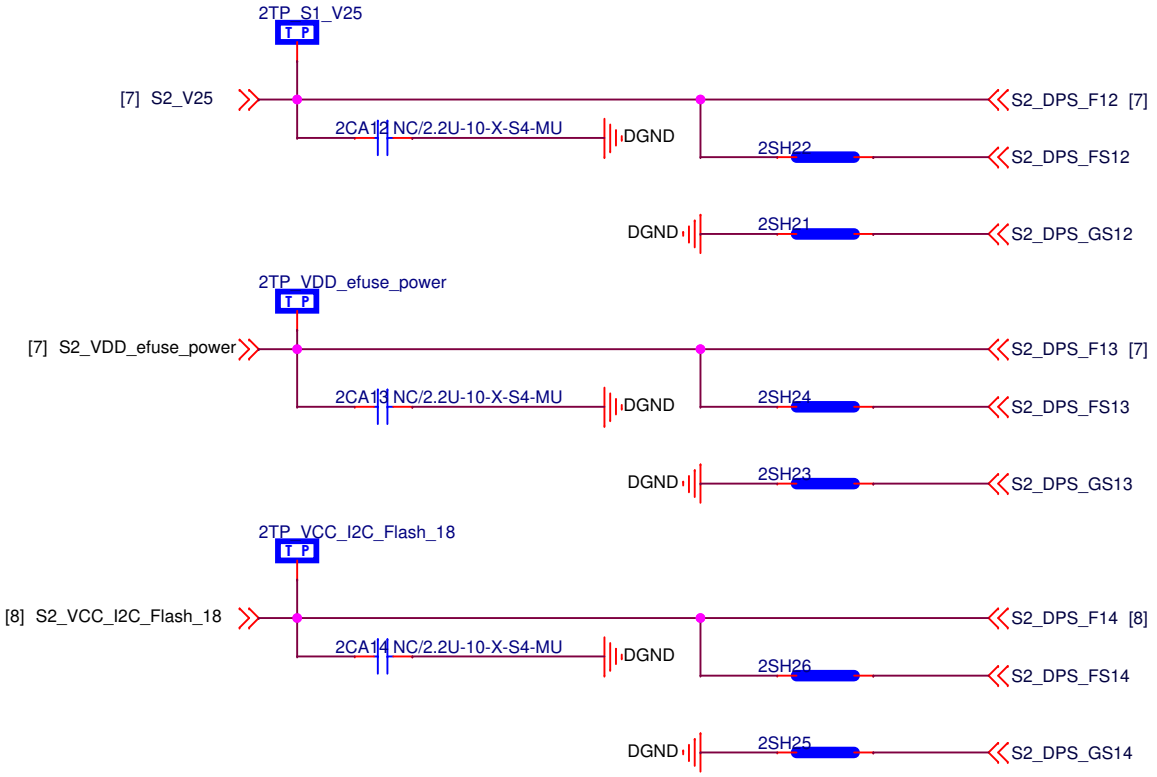


RLY close to ATE
C*--> Place on Bot ; Close RELAY



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	Rev. A	Size A2	PAGE 9 of 28

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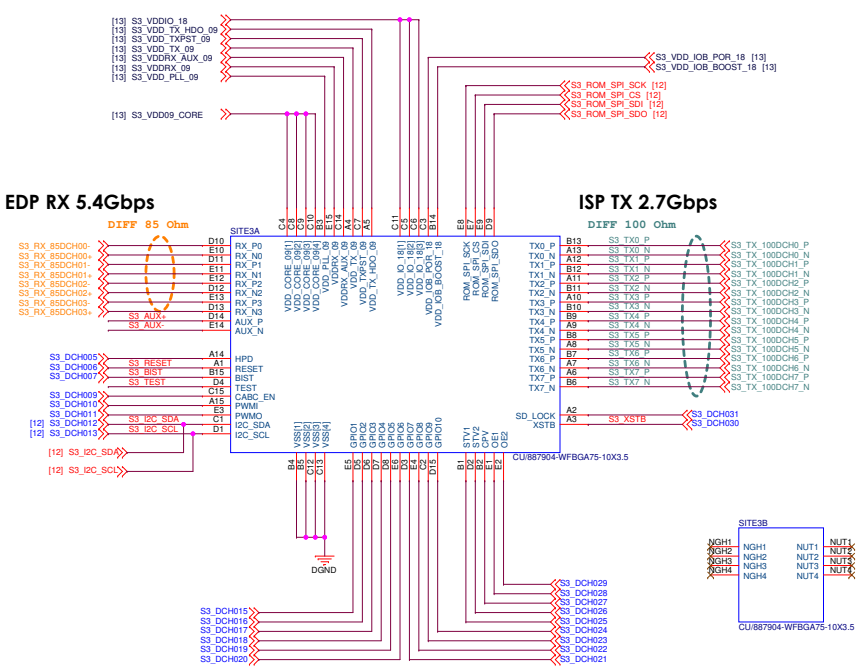
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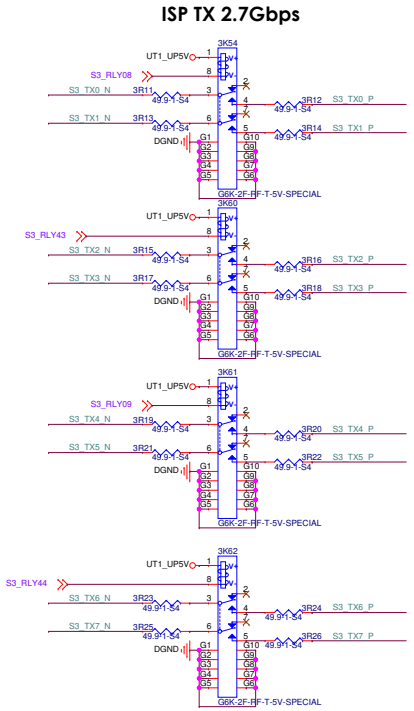
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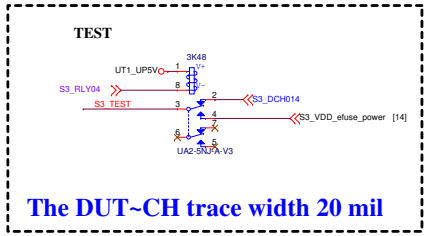
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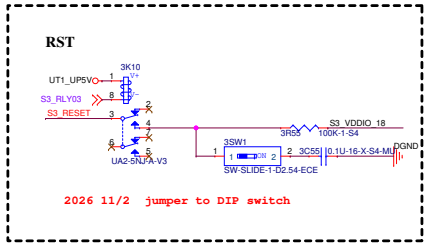
Rev.	Size	PAGE
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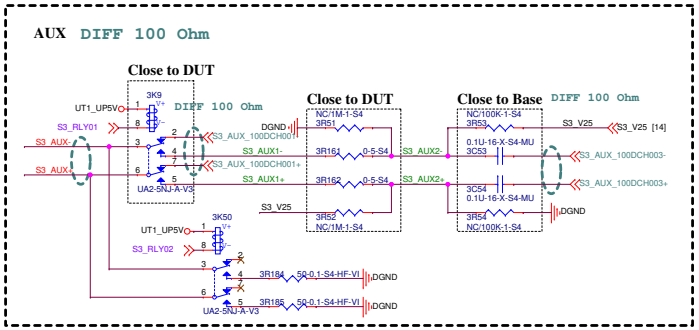
Swap P/N (P=+, N=-) to prevent RX routing from impacting trace length
IC PIN - RX_P0 connected to Test Channel - 85DCH-
IC PIN - RX_N0 connected toTest Channel - 85DCH+
IC PIN - RX_P1 connected to Test Channel - 85DCH-
IC PIN - RX_N1 connected to Test Channel - 85DCH+
IC PIN - RX_P2 connected to Test Channel - 85DCH-
IC PIN - RX_N2 connected to Test Channel - 85DCH+
IC PIN - RX_P3 connected to Test Channel - 85DCH-
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Rev.	Date		Modify History

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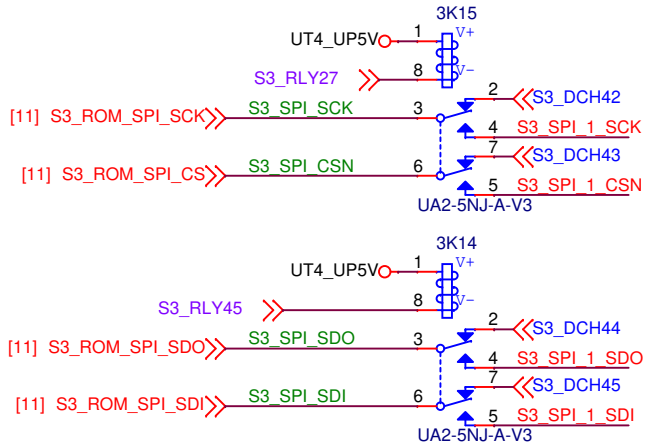
KEYSTONE MICROTECH
TEL:+886-3-5509980
FAX:+886-3-5501880
E-Mail:Service@KSMT.COM.TW

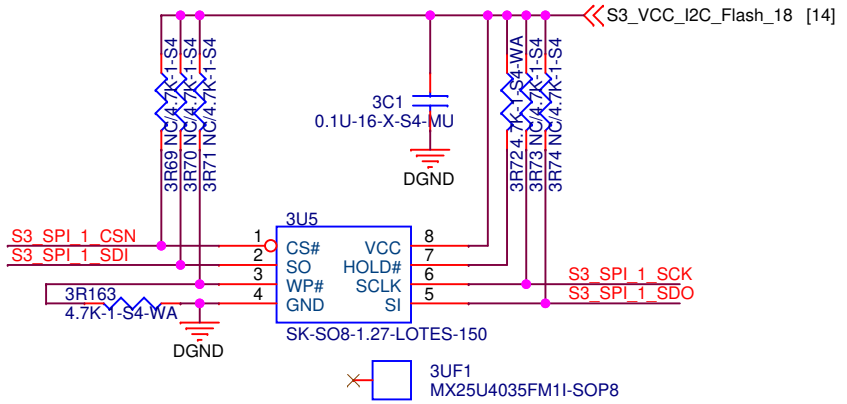
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PCB Number	038HX-887904K4Q-A100
Designer:	JACK Hsu
Ext.Num.:	8829
Approver:	<Approver>
Ext.Num.:	<Ext.>
Date:	Wednesday, November 11, 2026
Rev.	A
Size	A2
PAGE	11 of 28

FLASH

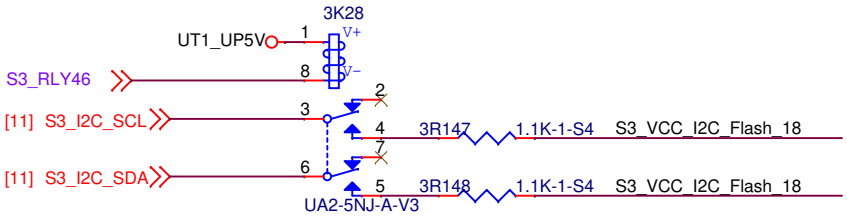
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※Please note that Flash uses the shortest path and all sites have the same length.
**Line length < 4000MIL





I2C

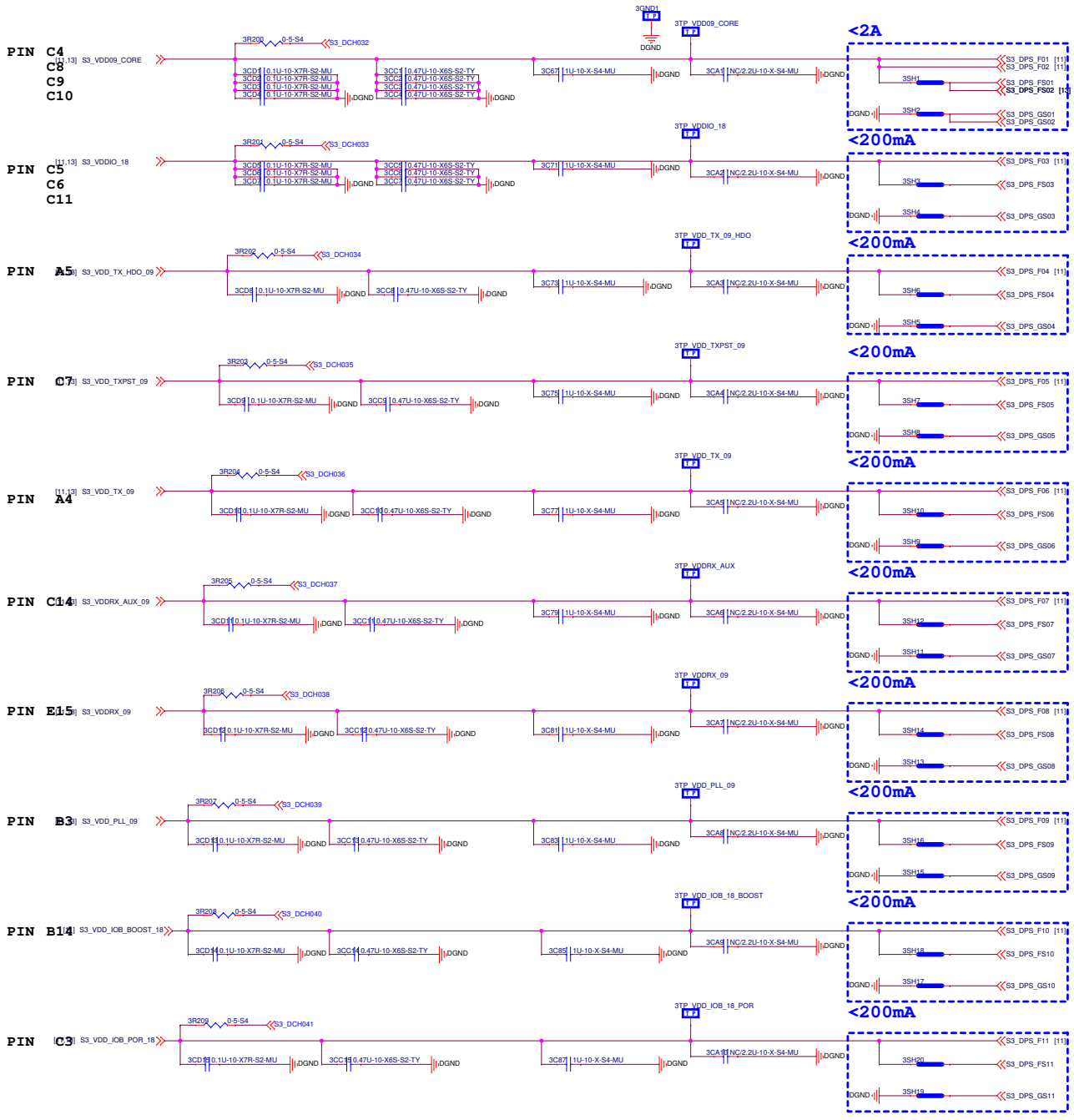


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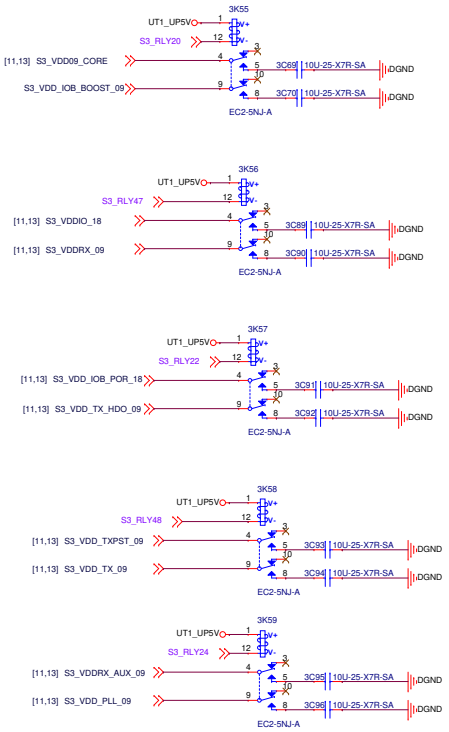
KEYSTONE MICROTECH
TEL:+886-3-5509980
FAX:+886-3-5501880
E-Mail:Service@KSMT.COM.TW

Page Name	P0032_S3_FLASH_I2C
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PCB Number	038HX-887904K4Q-A100
Designer:	JACK Hsu
Ext.Num.:	6629
Approver:	<Approver>
Ext.Num.:	<Ext.>
Date:	Wednesday, November 11, 2026
Rev.	A
Size	A4
PAGE	12 of 28

CAP Reference :
CD*-->Place on TOP ; close to PIN
CC*-->Place on TOP ; close to PIN
CA*-->Place on TOP ; close to ATE



RLY close to ATE
C*--> Place on Bot ; Close RELAY



Himax

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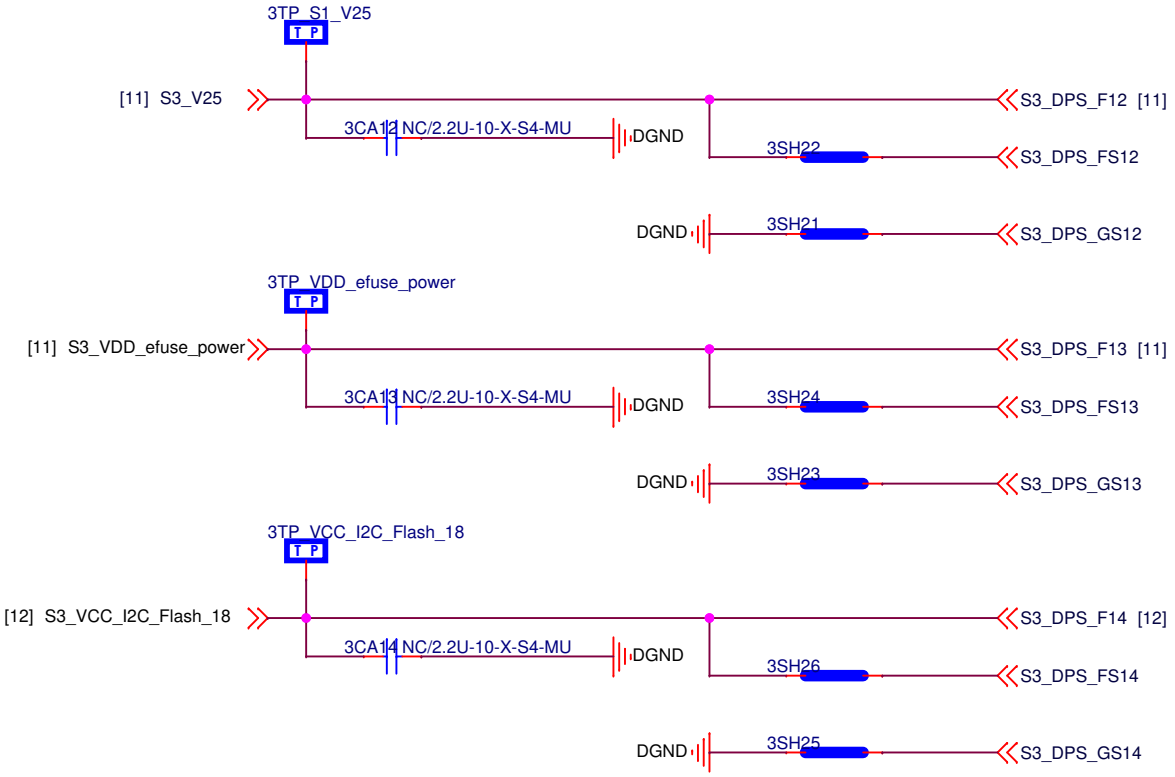
KEYSTONE MICROTECH
TEL:+886-3-5509880
FAX:+886-3-5501880
E-Mail:Service@KSMT.COM.TW

Page Name: P0033_S3_POWER
Product Name: 980HX-887904-KAG-A10
PCB Number: 038HX-887904K4Q-A100

Designer: JACK Hsu Ext.Num.: 6829
Approver: <Approver> Ext.Num.: <Ext.>
Date: Wednesday, November 11, 2026

Rev.	Size	PAGE
A	A2	13 of 28

CAP Reference :
CD*-->Place on TOP ; close to PIN
CC*-->Place on TOP ; close to PIN
CA*-->Place on TOP ; close to ATE



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Rev.	Date	Modify History

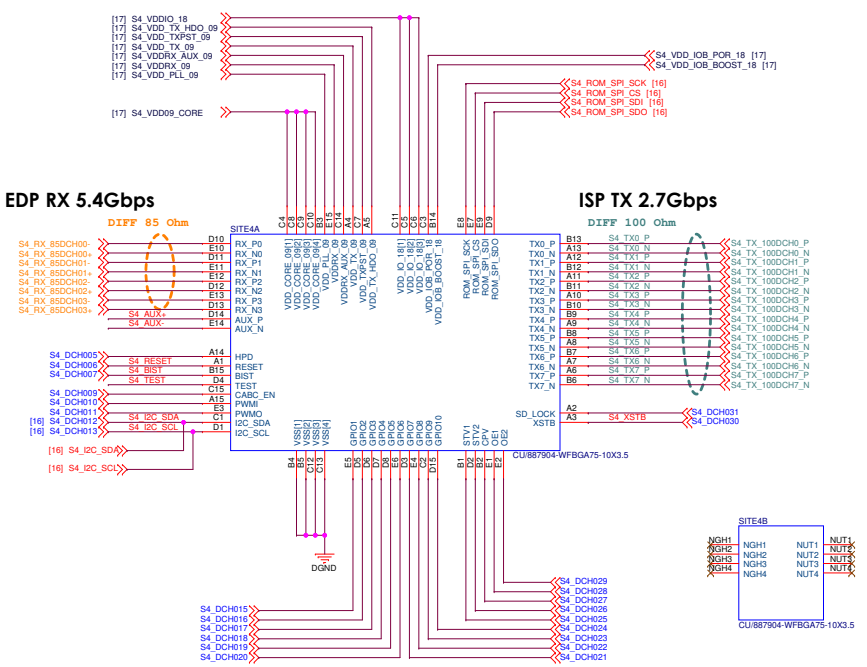
KEYSTONE MICROTECH
TEL:+886-3-5509980
FAX:+886-3-5501880
E-Mail:Service@KSMT.COM.TW

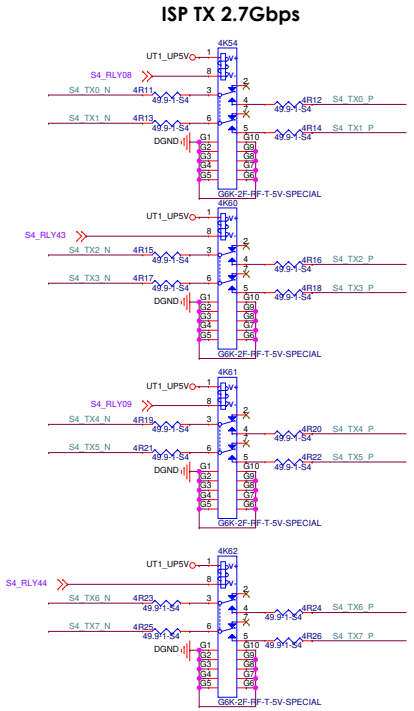
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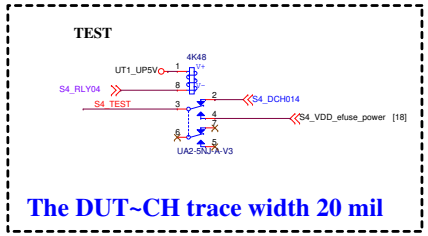
Designer: JACK Hsu
Ext.Num.: 6629
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Date: Wednesday, November 11, 2026

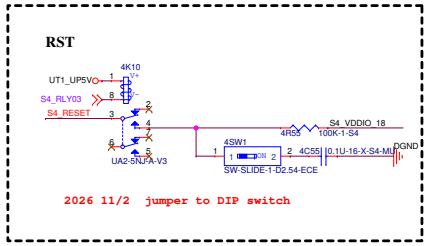
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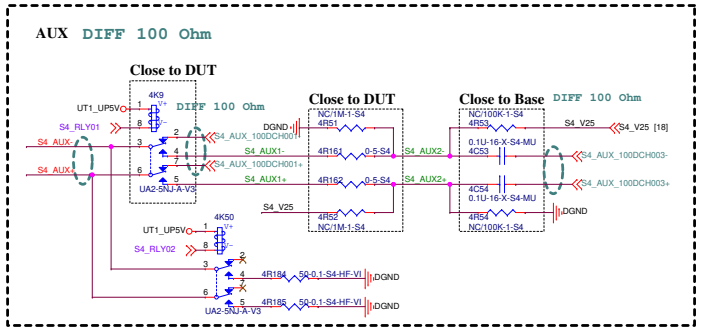
Swap P/N (P=+, N=-) to prevent RX routing from impacting trace length
IC PIN - RX_P0 connected to Test Channel - 85DCH-
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IC PIN - RX_N1 connected to Test Channel - 85DCH+
IC PIN - RX_P2 connected to Test Channel - 85DCH-
IC PIN - RX_N2 connected to Test Channel - 85DCH+
IC PIN - RX_P3 connected to Test Channel - 85DCH-
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Rev.	Date		Modify History

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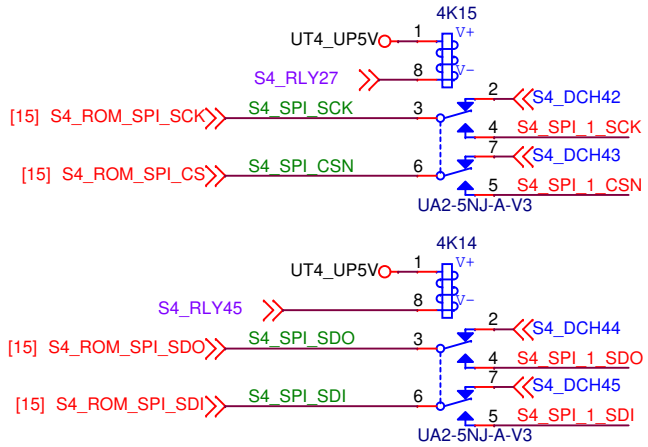
KEYSTONE MICROTECH
TEL:+886-3-5509980
FAX:+886-3-5501880
E-Mail:Service@KSMT.COM.TW

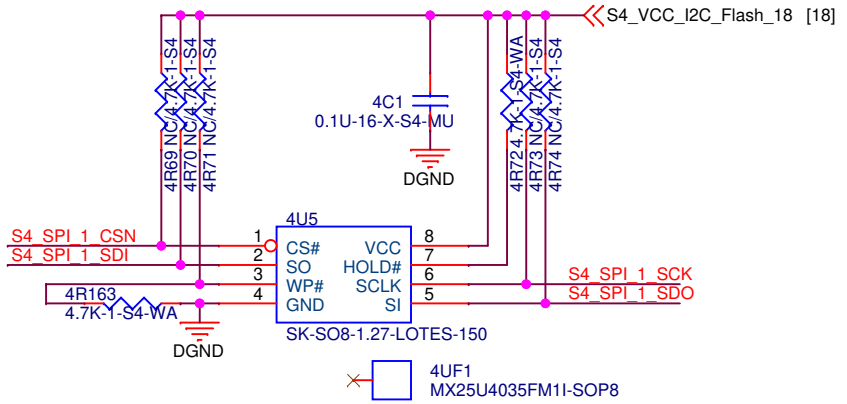
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Designer:	JACK Hsu
Ext.Num.:	8829
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Date:	Wednesday, November 11, 2026
Rev.	A
Size	A2
PAGE	15 of 28

FLASH

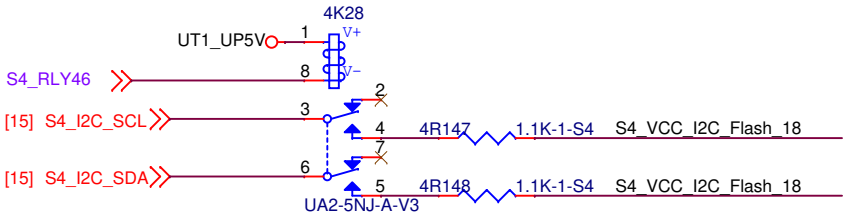
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※Please note that Flash uses the shortest path and all sites have the same length.
**Line length < 4000MIL





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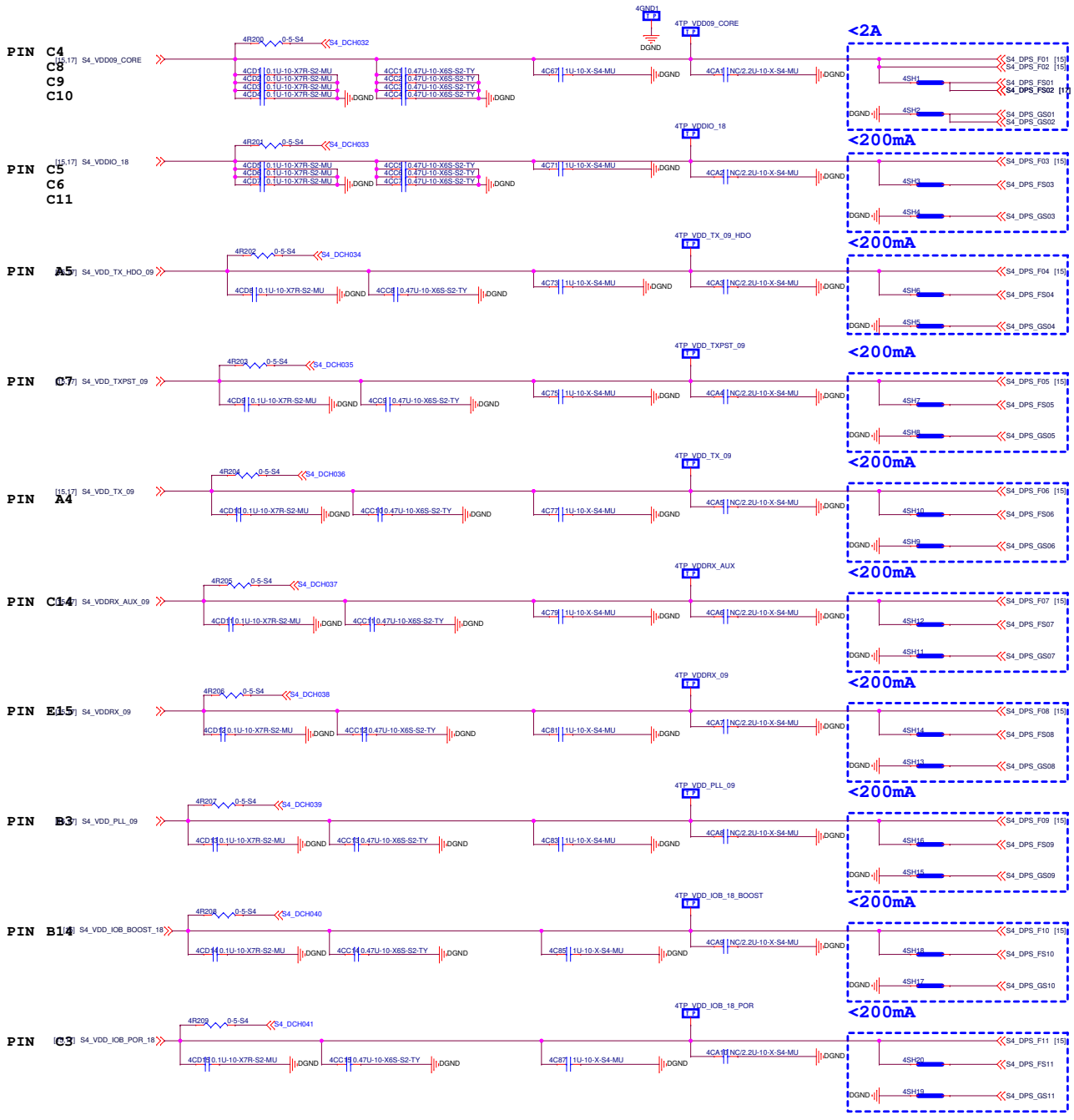
KEYSTONE MICROTECH
TEL:+886-3-5509980
FAX:+886-3-5501880
E-Mail:Service@KSMT.COM.TW

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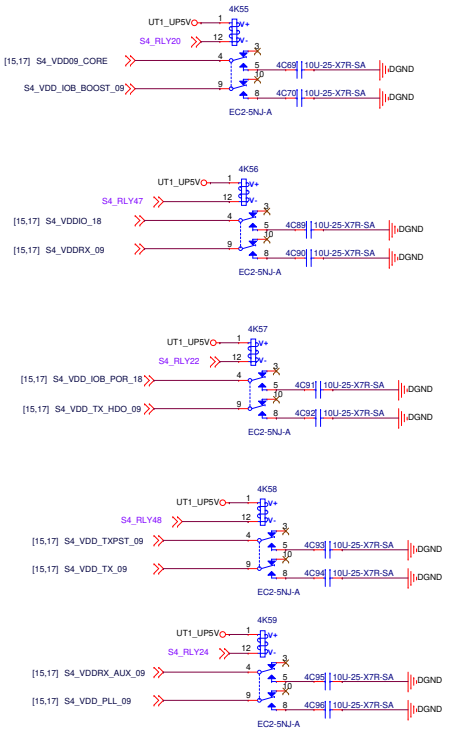
Designer: JACK Hsu
Ext.Num.: 6629
Approver: <Approver>
Ext.Num.: <Ext.>
Date: Wednesday, November 11, 2026

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CAP Reference :
CD*-->Place on TOP ; close to PIN
CC*-->Place on TOP ; close to PIN
CA*-->Place on TOP ; close to ATE



RLY close to ATE
C*--> Place on Bot ; Close RELAY



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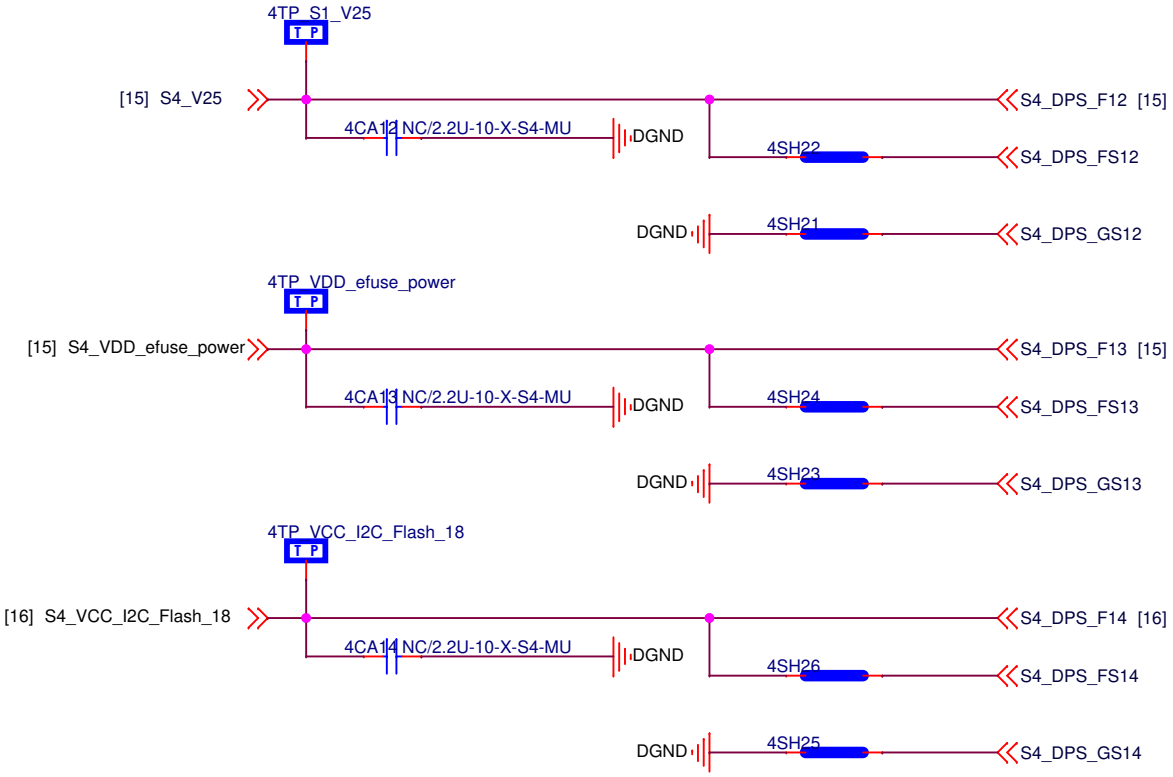
KEYSTONE MICROTECH
TEL:+886-3-5509880
FAX:+886-3-5501880
E-Mail:Service@KSMT.COM.TW

Page Name: P0043_S4_POWER
Product Name: 980HX-887904-KAG-A10
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Designer: JACK Hsu Ext.Num.: 6829
Approver: <Approver> Ext.Num.: <Ext.>
Date: Wednesday, November 11, 2026

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CAP Reference :
CD*-->Place on TOP ; close to PIN
CC*-->Place on TOP ; close to PIN
CA*-->Place on TOP ; close to ATE



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KEYSTONE MICROTECH
TEL:+886-3-5509980
FAX:+886-3-5501880
E-Mail:Service@KSMT.COM.TW

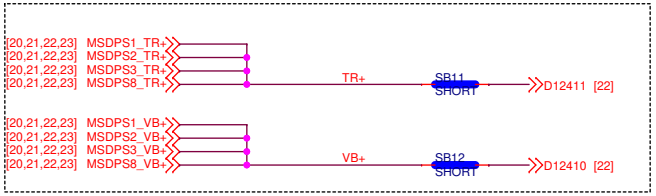
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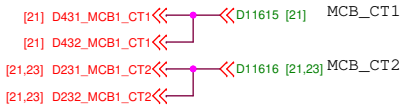
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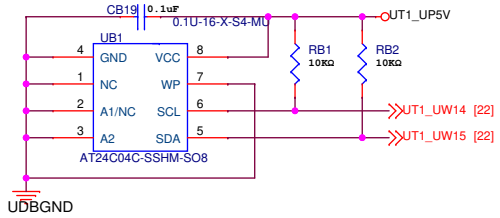
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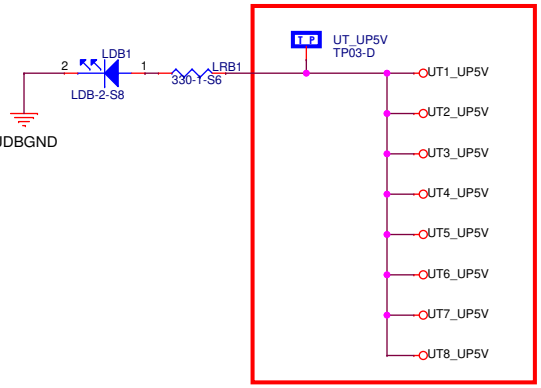
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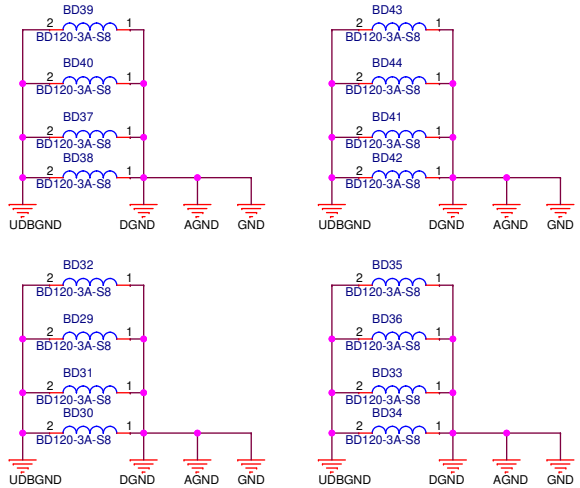


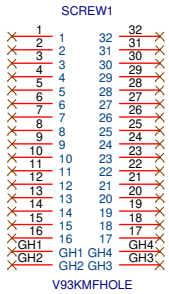






All UT_5V Merge





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	Rev.	Date	Modify History
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FAX:+886-3-5501880
E-Mail:Service@KSMT.COM.TW

Page Name	S9999_MISC	Designer:	JACK Hsu
		Ext.Num.:	6629
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PCB Number	038HX-887904K4Q-A100	Rev.	A
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		PAGE	19 of 28